

DESIGN AND ANALYSIS OF ALGORITHMS

Name: Haresh Kumar N L

Reg No: 192425009

Experiment 7: Quick Sort

Code:

```
def quick_sort(arr):  
    if len(arr) <= 1:  
        return arr  
    pivot = arr[-1]  
    left = [x for x in arr[:-1] if x <= pivot]  
    right = [x for x in arr[:-1] if x > pivot]  
    return quick_sort(left) + [pivot] + quick_sort(right)  
arr = [10, 7, 8, 9, 1, 5]  
sorted_arr = quick_sort(arr)  
print("Sorted Array:", sorted_arr)
```

Output:

```
===== RESTART: C:/Users/hk838  
Sorted Array: [1, 5, 7, 8, 9, 10]
```